When does multivariate forecast reconciliation help?

Supplementary material

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Experiment 1: alternative test statistics

The test of condition (d) in the main text uses Wilks' Λ with Rao's F approximation. Wilks' Λ is one of several standard statistics for a linear hypothesis in a multivariate regression, and here we compare it with two others. For variable j, let $\boldsymbol{E}$ and $\boldsymbol{E}_0$ be the residual sums of squares and cross-products matrices from the regression of its bottom-level errors on all the incoherences, and on its own incoherences only. Let $\boldsymbol{H} = \boldsymbol{E}_0 - \boldsymbol{E}$, and let $\lambda_1 \geq \cdots \geq \lambda_s$ be the non-zero eigenvalues of $\boldsymbol{E}^{-1}\boldsymbol{H}$. The three statistics are

$$\Lambda = \prod_{i=1}^{s} \frac{1}{1+\lambda_i}, \qquad V = \sum_{i=1}^{s} \frac{\lambda_i}{1+\lambda_i}, \qquad U = \sum_{i=1}^{s} \lambda_i,$$

namely Wilks' Λ, Pillai's trace V and the Hotelling–Lawley trace U. Each is converted to an approximate F statistic, using the same approximations as R's `anova()` for multivariate linear models. As in the main text, the tests for the two variables are combined with a Bonferroni correction.

Table S1 gives the rejection rates at the 5% level for the designs of Experiment 1 at $T = 100$ and $T = 200$. Pillai's trace and the Hotelling–Lawley trace behave almost exactly like Wilks' Λ. Under the separable and separable-plus-invisible covariances, their rejection rates over all four sample sizes and both hierarchies are 4.0–5.7% and 4.1–6.9% respectively. Their rejection rates never differ from those of Wilks' Λ by more than 3.5 percentage points, and by no more than 0.5 for $T \geq 500$. Pillai's trace is slightly conservative and the Hotelling–Lawley trace slightly liberal at the smallest sample sizes, with Wilks' Λ between the two. In the application, the p-values from Wilks' Λ, Pillai's trace and the Hotelling–Lawley trace are all < 0.001 for both base models, so the choice of statistic does not change the conclusions.

Table S1: Experiment 1: rejection rate at the 5% level of the test of condition (d) based on Wilks' Λ, Pillai's trace and the Hotelling–Lawley trace (H–L), by hierarchy, covariance design and sample size T (1000 replications). For F0 and F1, $\kappa = 0$ and the rejection rate is the size of the test.

		$T = 100$			$T = 200$		
Design	κ	Wilks	Pillai	H–L	Wilks	Pillai	H–L
Small hierarchy (8 series)							
F0	0.000	0.053	0.047	0.058	0.042	0.041	0.044
F1	0.000	0.050	0.048	0.055	0.042	0.041	0.045
F2, $\delta = 0.25$	0.056	0.066	0.064	0.069	0.101	0.101	0.104
F2, $\delta = 0.5$	0.106	0.171	0.159	0.178	0.358	0.348	0.365
F2, $\delta = 1$	0.179	0.648	0.633	0.661	0.964	0.964	0.965
F3, $\delta = 0.2$	0.084	0.142	0.122	0.150	0.297	0.288	0.303
F3, $\delta = 0.4$	0.171	0.642	0.633	0.652	0.976	0.976	0.976
F3, $\delta = 0.6$	0.270	0.996	0.996	0.996	1.000	1.000	1.000
Brazilian hierarchy (33 series)							
F0	0.000	0.050	0.046	0.064	0.044	0.043	0.049
F1	0.000	0.054	0.048	0.069	0.050	0.044	0.057
F2, $\delta = 0.25$	0.103	0.087	0.072	0.106	0.162	0.150	0.176
F2, $\delta = 0.5$	0.182	0.306	0.276	0.327	0.778	0.760	0.796
F2, $\delta = 1$	0.331	0.919	0.904	0.922	1.000	1.000	1.000
F3, $\delta = 0.2$	0.080	0.085	0.067	0.110	0.153	0.147	0.161
F3, $\delta = 0.4$	0.160	0.292	0.269	0.327	0.843	0.830	0.857
F3, $\delta = 0.6$	0.244	0.890	0.868	0.895	1.000	1.000	1.000

Experiment 2

Table S2 compares separate and joint MinT reconciliation with the base forecasts in Experiment 2. In every scenario, reconciliation reduces MSE by 6–13% relative to the ARIMA base forecasts and by 1–5% relative to the VAR base forecasts. The relative MSEs of separate and joint reconciliation never differ by more than 0.019.

Table S2: Experiment 2: MSE of separate and joint MinT reconciliation relative to the base forecasts, averaged over 500 replications, 16 series and horizons 1–12.

		ARIMA		VAR	
Scenario	T	Separate	Joint	Separate	Joint
Separable, $\rho = 0.7$	108	0.941	0.938	0.964	0.966
Separable, $\rho = 0.7$	400	0.905	0.886	0.981	0.980
Separable, $\rho = -0.7$	108	0.939	0.936	0.979	0.983
Separable, $\rho = -0.7$	400	0.896	0.884	0.991	0.990
Variable-specific dynamics	108	0.924	0.930	0.973	0.974
Variable-specific dynamics	400	0.889	0.885	0.977	0.974
Node-varying correlation	108	0.936	0.933	0.958	0.963
Node-varying correlation	400	0.893	0.885	0.982	0.980
Variable-specific node covariance	108	0.944	0.942	0.977	0.980
Variable-specific node covariance	400	0.913	0.905	0.982	0.983
Node-varying dynamics	108	0.939	0.938	0.960	0.960
Node-varying dynamics	400	0.888	0.875	0.959	0.954

Application: ETS base forecasts

Table S3 repeats the accuracy comparison of the main text with base forecasts from automatically selected ETS models. For ETS we estimate the covariance matrices from response residuals, because the innovation residuals of multiplicative-error models are relative errors.

Table S3: Application: MSE relative to base forecasts (ETS), geometric mean over the series at each level, averaged over 84 forecast origins (73 with all twelve horizons) and horizons 1–12.

	Admissions			Dismissals		
Method	Total	Regions	States	Total	Regions	States
Base	1.000	1.000	1.000	1.000	1.000	1.000
OLS	0.997	1.003	1.310	0.985	1.070	1.288
WLS (structural)	0.990	0.928	1.120	0.943	0.984	1.122
MinT, separate	0.949	0.883	0.996	0.877	0.902	0.964
MinT, separate (joint estimate)	0.958	0.889	1.000	0.880	0.905	0.966
MinT, joint	0.949	0.878	0.988	0.815	0.852	0.925

Application: results by horizon

Table S4 breaks down the comparison of joint and separate reconciliation (ARIMA base forecasts) by forecast horizon. For admissions the ratio is above one at every horizon and level, and falls towards one as the horizon grows. For dismissals it is below one at every horizon and level.

Table S4: Application (ARIMA base forecasts): MSE of joint relative to separate MinT reconciliation by forecast horizon, geometric mean over the series at each level.

	Admissions			Dismissals		
Horizon	Total	Regions	States	Total	Regions	States
1–3	1.059	1.045	1.034	0.912	0.923	0.957
4–6	1.042	1.033	1.029	0.884	0.890	0.925
7–9	1.033	1.023	1.025	0.910	0.901	0.927
10–12	1.017	1.010	1.013	0.936	0.921	0.939

Application: results for every series

Table S5 gives the results series by series. For dismissals, joint reconciliation is better than separate reconciliation for every series. For admissions it is better for only 8 of the 33 series, and the differences are small.

Table S5: Application (ARIMA base forecasts): MSE of separate and joint MinT reconciliation relative to the base forecasts for every series, averaged over 84 origins (73 with all twelve horizons) and horizons 1–12.

	Admissions		Dismissals	
Series	Separate	Joint	Separate	Joint
Total	0.952	0.981	1.420	1.302
Midwest	1.017	1.019	0.879	0.802
Northeast	1.005	1.053	0.906	0.790
North	0.914	0.911	1.430	1.266
Southeast	0.998	1.034	1.249	1.176
South	0.975	1.000	0.890	0.839
AC	0.997	1.019	1.125	1.109
AL	0.944	0.960	0.959	0.933
AM	1.012	0.991	1.297	1.086
AP	0.957	0.911	1.145	1.100
BA	0.979	0.985	1.223	1.131
CE	0.962	1.017	1.061	0.929
DF	0.940	0.938	1.185	1.144
ES	0.984	0.994	1.066	0.992
GO	0.972	0.957	1.082	0.980
MA	0.929	0.963	1.176	1.111
MG	0.974	0.997	1.015	0.946
MS	0.948	0.909	1.250	1.150
MT	0.890	0.960	1.071	1.003
PA	0.923	0.931	1.141	1.098
PB	0.929	0.953	1.012	0.974
PE	0.939	0.962	1.111	0.985
PI	0.894	1.006	1.256	1.190
PR	0.999	1.025	1.194	1.155
RJ	0.994	1.026	1.058	1.019
RN	0.903	1.017	1.213	1.115
RO	0.973	0.973	1.160	1.021
RR	0.896	0.922	1.045	1.010
RS	0.999	1.054	1.083	1.012
SC	0.995	0.985	1.068	0.993
SE	0.943	0.934	1.086	1.075
SP	0.983	1.023	0.943	0.899
TO	0.837	0.864	1.244	1.166